

Identifying Fraud in Electricity and Gas Consumption

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National University Singapore (NUS) MSI5001: Introduction to AI: Concepts, Applications, and Evaluation

Introduction

Energy theft and fraudulent use of electricity or gas have become major challenges for utility companies worldwide. This issue goes beyond financial loss — it also affects the reliability of energy supply and the fairness of access for all users. We selected this topic because it represents a real-world situation where artificial intelligence can effectively enhance operational efficiency. Manually investigating potential fraud cases is time consuming and resource-demanding, therefore developing an automated model helps utility companies to identify suspicious accounts more quickly and reduce dependence on manual checks.

In this project, we explored a wide range of machine learning models to identify fraudulent activity and find the best approach that balances AUC, accuracy, precision, and recall. We also focused on keeping the model easy to understand and scalable, ensuring it can reliably detect fraud while reducing missed cases.

Methodology

This study utilized the *Fraud in Electricity and Gas Consumption* dataset, which contains client-level and invoice-level information from over 11,000 customers. The dataset is highly imbalanced, with fraudulent cases forming a very small proportion of the total. This project was framed as a binary classification task, where the goal was to distinguish between fraudulent and legitimate clients based on their billing and consumption patterns.

Data Preprocessing

We began by cleaning and organizing the datasets to ensure accuracy and consistency. We examined missing values, corrected inconsistencies, and **merged electricity and gas records to maintain data integrity**. We then performed **exploratory data analysis (EDA)** to discover consumption patterns and identify anomalies. We also **understand data relationships and the fraud distribution of the data**. We used **visualizations created with matplotlib and seaborn** to better understand the class imbalance and overall consumption patterns.

To improve model learning, we first conducted a **correlational analysis** to examine relationships between features before applying **outlier detection** to handle extreme values. We also used **Winsorization(cap at 99th percentile)** to cap extreme values, helping to reduce the

impact of outliers and keep the data more balanced and reliable for model training.

In addition, we performed **feature engineering and aggregation** to create new indicators such as irregular billing and usage variance. Although we tried removing irrelevant or weakly correlated information, it did not improve performance, so we kept all features. We then applied **One-Hot Encoding to convert categorical data** and used **Robust Scaler for feature scaling** to manage skewed distributions. The data was split using perfect stratification to maintain class balance, and **SMOTE was applied to address data imbalance**. These systematic steps ensured the dataset was reliable, balanced, and well-structured for effective model training and evaluation.

Model Selection

We framed fraud detection as a **binary classification problem** to separate fraudulent from legitimate customers. Starting with **Logistic Regression** as a baseline, we applied **Random Forest** with class weighting to capture non-linear patterns and manage data imbalance more effectively across the dataset. We enhanced detection sensitivity using **XGBoost** with `scale_pos_weight` and performed hyperparameter tuning to refine key parameters. We also tested **LightGBM** for faster training and a **LightGBM + Isolation Forest hybrid model** to detect rare, atypical fraud cases. Additionally, we used **SMOTE with Random Forest** to improve recall on minority classes.

Across all models, we conducted **Randomized Search** and **Grid Search** for optimization. Performance was measured using **AUC, precision, recall, and F1-score**, with AUC as the main metric due to its robustness in imbalanced data. This ensured balanced, accurate, and stable fraud detection performance.

Model Validation

In this stage, we use **stratified k-fold cross-validation** to ensure that the proportion of fraud and non-fraud cases remains consistent across all folds. This process can increase the reliability and consistency of the model. **Hyperparameter tuning** was performed to find the best settings that improve model performance and prevent overfitting. We applied **threshold tuning using precision-recall analysis** to find the most suitable decision point, allowing us to balance between detecting more fraud cases and minimizing false alarms based on operational needs.

Model Evaluation

We compared different models using the same validation steps to ensure fair and reliable evaluation. Instead of just looking at the final results, we focused on how each model learned from the imbalanced data. We used several metrics—AUC, precision, recall, and F1-score—to measure different parts of model performance. Among these, **AUC was our main focus** because it shows how well the model separates fraud from non-fraud cases across all thresholds, which is especially important for imbalanced datasets. We also examined **feature importance** and **decision tree splits** to understand which factors influenced fraud detection most, keeping the process clear and easy to explain. We also tried an **ensemble approach**, combining multiple models to leverage their strengths, which resulted in improved overall performance and more stable predictions.

Results

Top 5 Models

#	Model	Threshold	Precision	Recall	F1	AUC
1	Voting Ensemble	0.35	0.112	0.650	0.191	0.792
2	RF Balanced (tuned)	0.34	0.130	0.553	0.211	0.788
3	RF GridSearch	0.43	0.130	0.563	0.211	0.784
4	RF RandomSearch	0.33	0.119	0.621	0.199	0.783
5	XGBoost (Tuned)	0.37	0.107	0.660	0.184	0.777

Table 1: Top 5 Performing Models and their Results sorted by AUC Score

Best Performing Model

Table 1 above summarizes the results of the top 5 models we used on the dataset based on AUC score. The best performing mode, and therefore the model we chose is **Voting Ensemble** (combining Random Forest, XGBoost, and Logistic Regression) with a tuned decision threshold of 0.35. It achieved the best AUC (0.792) amongst the models we tried, while performing well for Recall (0.650 - caught 65.0% of frauds in the test dataset) with an F1-score of 0.191.

The ensemble approach leverages the complementary strengths of multiple algorithms, resulting in more robust fraud detection. Notably, RF Balanced (AUC=0.788, F1=0.211) serves as a strong alternative with better precision-recall balance. The resulting investigation workload was 597 customers (25.4% of the test set) with 530 false positives ($\approx 22.6\%$ false-alarm rate) - a manageable level for operational deployment. In business terms, targeting the Critical & High risk tiers (probability ≥ 0.35) captures 65.0% of total frauds by reviewing only

25.4% of customers, delivering an estimated ROI of 124%, with a net benefit of \$185,200 when scaled to the full customer base of 11,710.

Comparison with other Models

RF Balanced (AUC=0.788, F1=0.211) achieved the best precision-recall balance among individual models, with higher precision (0.130 vs 0.112) than the ensemble but lower recall (55.3% vs 65.0%), resulting in 381 false positives - a 28% reduction in investigation workload for precision-focused operations.

RF GridSearch and **RandomSearch** (AUC=0.784, 0.783) demonstrated marginal gains from hyperparameter tuning, with GridSearch matching RF Balanced's F1 (0.211) while RandomSearch achieved higher recall (0.621) at lower precision (0.119).

XGBoost Tuned (AUC=0.777) improved significantly from baseline (+0.032 AUC) with strong recall (0.660) but low precision (0.107).

Conclusion

Our evaluation shows that ensemble methods outperform individual classifiers in detecting fraudulent clients. The **Voting Ensemble** (Random Forest + XGBoost + Logistic Regression) achieved the highest discrimination ability (AUC=0.792), demonstrating that aggregating diverse model perspectives captures fraud patterns more robustly than any single approach.

While Logistic Regression served as a useful benchmark, its linear assumptions struggled with irregular usage patterns characterizing fraud. The ensemble and Random Forest variants captured these complexities, surfacing a clear risk signature: unstable Tier-2/Tier-3 consumption (oscillating usage) and inconsistent meter coefficients emerged as strongest predictors. This pairing of behavioral volatility and technical anomalies provides targets utilities can monitor automatically.

Beyond model lift, the approach delivers actionable intelligence. The five-tier risk classification translates probabilities into prioritized workflows, enabling teams to focus audits where recoveries are most likely capturing 65% of frauds by investigating only 25% of customers.

We recommend adopting the Voting Ensemble as the primary detection model, with RF Balanced as a precision-focused alternative, operated under business-owned threshold governance to flex recall vs. workload as conditions evolve. This balances effectiveness (124% ROI, \$185K net benefit) with operational control, ensuring the system remains accurate, explainable, and economically sound over time.

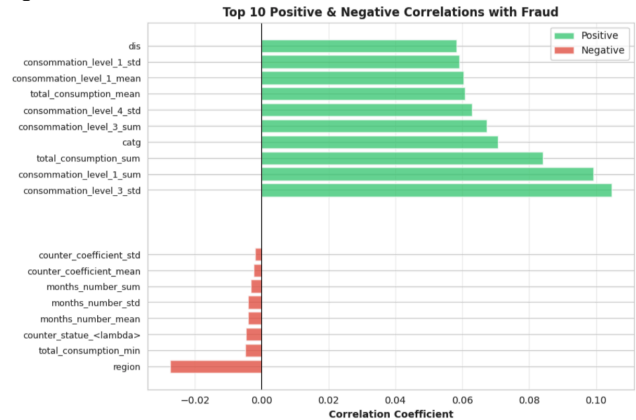
Appendix

Data Preprocessing:

Having a clean and well-prepared dataset is crucial before choosing any model. It helps the model learn real and meaningful patterns instead of random noise, reduces bias from data imbalance or errors, and improves accuracy and consistency. By creating 110 detailed and thoughtful features, we strengthened the model's ability to detect subtle signs of fraud while keeping the data organized, easy to understand, and ready for reliable training.

Step	Method Used	Why It Is Important
1. Data Cleaning and Integration	Checked for missing values, corrected inconsistencies, and merged electricity and gas datasets.	Ensures data accuracy, completeness, and consistency across all records, reducing potential model bias.
2. Exploratory Data Analysis (EDA)	Used matplotlib and seaborn visualizations to explore consumption patterns, fraud distribution, and class imbalance.	Helps us understand data relationships and identify irregularities or biases that may affect model training.
3. Correlational Analysis	Examined relationships among variables before applying outlier handling.	Identifies redundant or weakly correlated variables, supporting informed feature selection and interpretation.
4. Outlier Detection and Treatment	Applied Winsorization at the 99th percentile and IQR / z-score methods to handle extreme values.	Caps extreme outliers while preserving key data trends, preventing distortions during model learning.
5. Feature Engineering (110 Features from 17 Original)	Created new behavioral, temporal, and statistical indicators to capture complex consumption and billing patterns.	Improves predictive accuracy by incorporating detailed usage, billing, and time-based behavioral data.
5.1 Aggregation Strategy (Transaction → Customer Level)	Aggregated invoice-level data into customer-level summaries.	Ensures each customer is represented by meaningful summarized information suitable for classification.
5.2 Statistical Aggregates (40 features)	Mean, median, std, min, max, sum, range for consumption and billing periods.	Captures variability and central tendencies that may signal fraudulent consumption.
5.3 Consumption Patterns (15 features)	Coefficient of variation, trend direction, peak-to-average ratio, zero-consumption counts.	Highlights irregular or inconsistent usage behaviors linked to fraud.
5.4 Level Usage (25 features)	Ratios between consumption levels, average levels used, and zero-level counts.	Detects abnormal use of higher consumption tiers commonly observed in fraud cases.
5.5 Billing Patterns (12 features)	Average days between invoices, billing frequency variation, short/long billing cycle proportions.	Reveals inconsistent billing activity that may suggest tampering or irregularities.
5.5 Temporal Features (15 features)	Customer lifespan, registration era, and seasonal usage metrics.	Accounts for time-related trends and behavioral shifts over a customer's history.
5.6 Categorical Encoding (5 features)	Created high-risk category flags (e.g., is_category_51) and one-hot encoding for region/district.	Converts categorical variables into numerical format while keeping risk information intact.
6. Feature Scaling	Applied RobustScaler (resistant to outliers, uses median/IQR).	Standardizes numeric features while minimizing the effect of outliers, ensuring balanced model input.
7. Stratified Train-Test Split	Split dataset 80/20 , maintaining 4.39% fraud rate in both sets through stratification .	Preserves class proportions between training and test data, enabling fair model evaluation.
8. Data Balancing	Applied SMOTE (Synthetic Minority Oversampling Technique) .	Addresses data imbalance by generating synthetic minority samples, improving recall and detection performance.

Top Correlated Features:



Top Positive and Negatively correlated Features versus Target = 1 (Positive Cases). With the highest correlation coefficient being just 0.1, it is that there is no clear relationship between the Features and the Target.

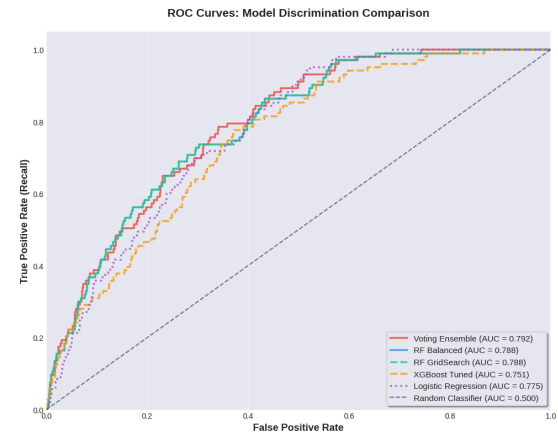
Multicollinearity in dataset:

Found 11 highly correlated feature pairs ($|r| > 0.8$):

	Feature 1	Feature 2	Correlation
1	total_consumption_std	total_consumption_max	0.970
9	months_number_mean	months_number_sum	0.958
6	consummation_level_3_mean	consummation_level_3_sum	0.936
3	total_consumption_max	consummation_level_2_std	0.910
10	months_number_mean	months_number_std	0.908
4	consummation_level_2_mean	consummation_level_2_sum	0.886
8	consummation_level_4_mean	consummation_level_4_sum	0.870
2	total_consumption_std	consummation_level_2_std	0.866
5	consummation_level_2_std	consummation_level_2_sum	0.858
7	consummation_level_4_mean	consummation_level_4_std	0.855
0	total_consumption_mean	total_consumption_sum	0.849

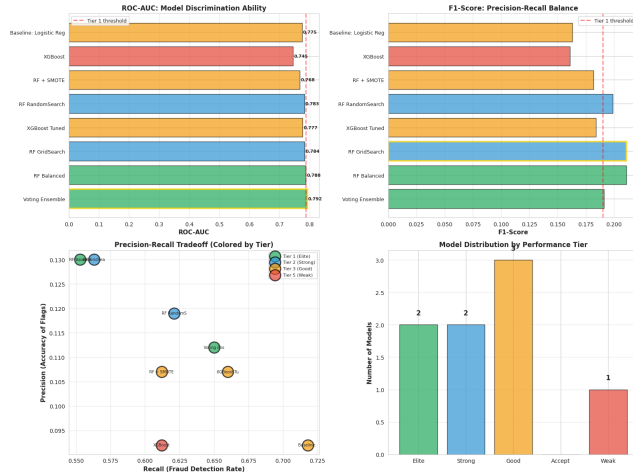
Examples of highly correlated feature pairs (multicollinearity), which can cause bias in the data if not processed out

AUC plot of Top 6 models:



AUC plot, showing **Voting Ensemble** (combining Random Forest, XGBoost, and Logistic Regression) having the largest variance from the random predictor $y=x$ line

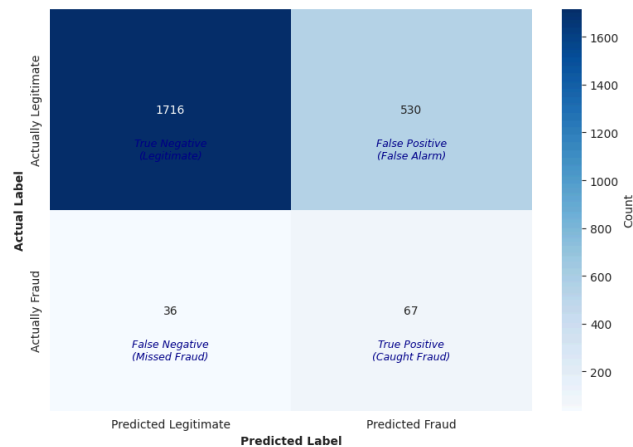
Model Comparison based on Precision-Recall Tradeoff:



Voting Ensemble (combining Random Forest, XGBoost, and Logistic Regression) showing the most balanced result when testing for Recall and Precision on the test data. Priority was to aim for a higher Recall, than Precision given the importance of flagging actual fraud cases (True Positive) versus wrongly flagging non fraud cases (False Positive)

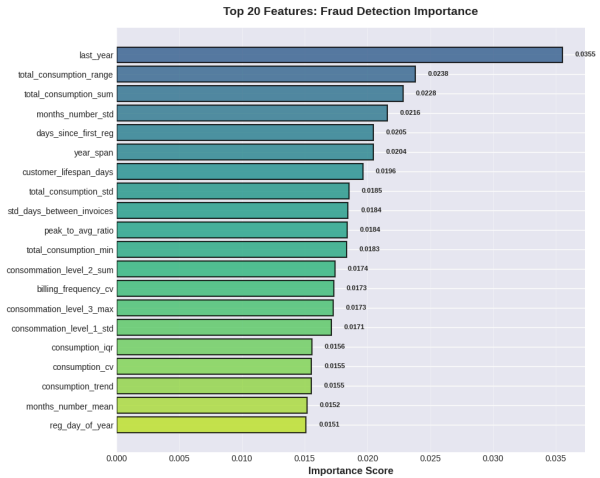
Confusion Matrix of Voting Ensemble:

Confusion Matrix - Voting Ensemble (t=0.35)



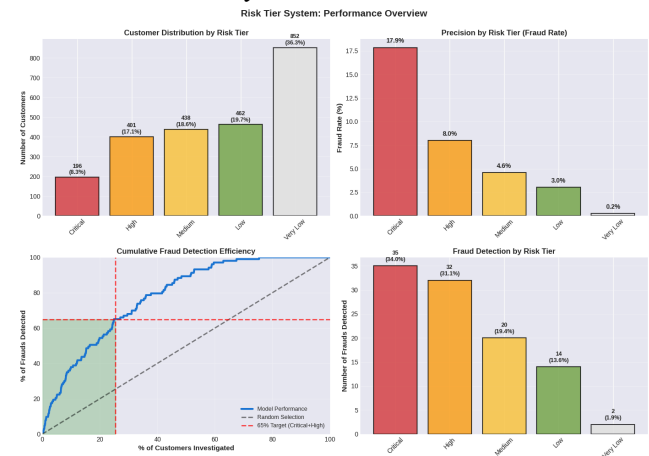
This confusion matrix for the Voting Ensemble modal (threshold = 0.35) shows that 65% of Fraud cases were caught (True Positive), while 35% of Fraud cases were missed out (False Negative)

Top 20 Feature Importance:



This chart shows the top 20 features in terms of their importance at towards assisting the modal in predicting the target variable (fraud or not fraud)

Customer Risk Tier System:



Customers were grouped into 5 Risk Tiers based on the probability of them being a risk. For example, Probability ≥ 0.5 means 'Critical', Probability ≥ 0.35 means 'High'

Acknowledgement of AI tool usage:

We acknowledge the use of **AI-assisted tools**, including **ChatGPT** and **Claude**, to support our project development. These tools were used responsibly to **enhance learning, refine Python scripts, and clarify methodological concepts**, based on our own inputs and understanding. All analytical decisions, interpretations, and final implementations were made independently by our team. The AI tools served solely as aids for **code debugging, documentation structuring, and idea validation**, ensuring our workflow remained transparent, accurate, and educationally driven.